FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Main)-2024

TEST DATE: 27-03-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. D

Sol. Potential energy of system decreases in spontantenous process

$\Rightarrow a_{com}$ downward

$mg - N = ma$

$\Rightarrow N = m(g - a)$

$N < mg$

2. A

Sol. $\left|\vec{\tau}_P\right| = r_\perp F = \dfrac{\left|3(2) - 4\times(1) - 5\right|}{5} \times 30 = 18$

Since point 'P' lies on the side of origin, $\vec{\tau}_P = 18(-\hat{k})$ & point 'Q' lies on the line $\Rightarrow \vec{\tau}_Q = \vec{0}$

3. C

Sol. $(6i + \alpha\hat{k}) \cdot (-3\hat{i} + 2\hat{j} + 2\hat{k})$

$-18 + 2\alpha = 0$

$\Rightarrow \alpha = 9$

4. A

Sol. $F_{net} = q(\vec{E} + \vec{v} \times \vec{B}) = \vec{0}$

$\vec{E} = -\left|\vec{v} \times \vec{B}\right|$

$\vec{E} \perp \vec{v}$

$\vec{E} \perp \vec{B}$

5. C

Sol. $\frac{mv^2}{R} = \frac{GMm}{R^{3/2}}$

$V = \sqrt{\frac{GM}{R^{1/2}}}$

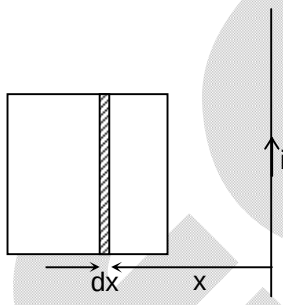
$T = \frac{2\pi R}{V} = \frac{2\pi R^{5/4}}{\sqrt{GM}}$

6. B

Sol. Apply principle of reciprocity

$\int d\phi = \int_b^{b+a} \frac{\mu_0 i}{2\pi x} a dx$

$\phi = \frac{\mu_0 i a}{2\pi} \ell n\left(\frac{a+b}{b}\right)$



7. C

Sol. $\vec{F}_{net} = \int d\vec{F} \odot = \int B_0 i d\ell \odot$

$\vec{F}_{net} = i(2\pi R)B_0 \odot$

8. A

Sol. Since, $P_i = 0 \Rightarrow P_f = 0$

$P_1 = P_2 = P$

$\Rightarrow \lambda_1 = \lambda_2$

$R = \frac{mv}{qB} = \frac{P}{q_1 B}$ …(i)

$2R = \frac{P}{q_2 B}$ …(ii)

From (i) and (ii),

$\frac{q_1}{q_2} = 2$

9. B

Sol. $13.6z^2\left[\frac{1}{2^2} - \frac{1}{n^2}\right] = 10.2 + 17 = 27.2$ …(i)

$13.6z^2\left[\frac{1}{3^2} - \frac{1}{n^2}\right] = 4.25 + 5.95 = 10.2$ …(ii)

On solving (i) and (ii),

n = 6, z = 3

10. B

Sol. Applying Gauss' law

$E \cdot 4\pi r^2 = \frac{1}{\varepsilon_0} \int_0^r \rho \cdot 4\pi r^2 dr$

$(Ar^2)(4\pi r^2) = \frac{1}{\varepsilon_0}\int_0^r \rho(4\pi)r^2 dr$

$Ar^4 = \frac{1}{\varepsilon_0}\int_0^r \rho r^2 dr$

Differentiate

$4Ar^3 = \frac{1}{\varepsilon_0}\rho r^2$

$\Rightarrow \rho = 4\varepsilon_0 Ar$

11. B

Sol. For isothermal process : $Q = W \Rightarrow \tan\theta = 1$

For adiabatic process : $Q = 0 \Rightarrow \frac{W}{Q} = \infty \Rightarrow \tan\theta \Rightarrow \infty$

For isobaric process:

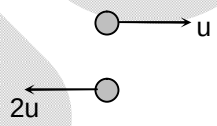
$Q = nC_P\Delta T$

$W = n(C_P - C_V)\Delta T$

$\tan\theta = \frac{W}{Q} = \frac{C_P - C_V}{C_P} = 1 - \frac{C_V}{C_P} = 1 - \frac{1}{\gamma}$

12. B

Sol. Change in momentum, I = m(3U)



$W = \Delta K.E. = \frac{1}{2}m\left[(2U)^2 - U^2\right]$

$= \frac{3}{2}mU^2 = \frac{IU}{2}$

13. C

Sol. Let velocity of COM of rod is v_C

$v_C - \frac{\omega L}{2} = u$

$v_C = u + \frac{\omega L}{2}$

$K.E. = \frac{1}{2}mv_C^2 + \frac{1}{2}I_C\omega^2$

$= \frac{1}{2}M\left(u + \frac{\omega L}{2}\right)^2 + \frac{1}{2}\frac{ML^2}{12}(\omega)^2$

$= \frac{1}{2}Mu^2 + \frac{1}{6}ML^2\omega^2 + \frac{1}{2}ML\omega u$

14. B

Sol. $T = \frac{2\pi R}{u} \Rightarrow u = \frac{2\pi R}{T}$

Range $= \frac{u^2 \sin 2\theta}{g} = 2R$

$\theta = \frac{1}{2}\sin^{-1}\left(\frac{2gR}{u^2}\right) = \frac{1}{2}\sin^{-1}\left(\frac{2gRT^2}{4\pi^2R^2}\right) = \frac{1}{2}\sin^{-1}\left(\frac{gT^2}{2\pi^2R}\right)$

15. D

Sol. $P = 10\ mW = 5I_Z$

$I_Z = 2\ mA$

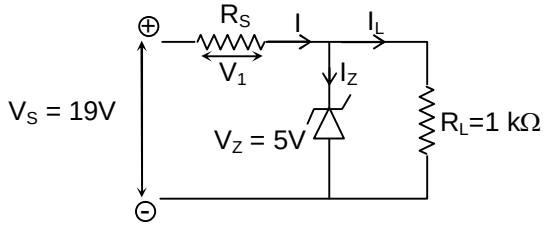
$I_L = \dfrac{5}{1\times10^3} = 5mA$

$I = 2 + 5 = 7\ mA$

$V_1 = 19 - 5 = 14\ V$

$I = \dfrac{V_1}{R_S}$

$R_S = \dfrac{V_1}{I} = \dfrac{14}{7mA} = 2000\,\Omega$



16. C

Sol. $\dfrac{B}{KT}$ is dimensionless

$[B] = [KT]$

Also, $[I] = \left[AT^2\right]$

$[A] = [IT^{-2}]$

$\therefore\ [AB^2] = [IT^{-2}K^2T^2] = [IK^2]$

17. B

Sol. $\vec{E} = \dfrac{1}{2}E_0\hat{i}\left[2\cos(kz)\cos(\omega t)\right]$

$2\cos C\cos D = \cos(C+D) + \cos(C-D)$

$\vec{E} = \dfrac{E_0}{2}\left[\cos(kz+\omega t)\hat{i} + \cos(kz-\omega t)\hat{i}\right]$

$\vec{B} = \dfrac{E_0}{2C}\left[\cos(kz+\omega t)(-\hat{j}) + \cos(kz-\omega t)\hat{j}\right]$

$\vec{B} = \dfrac{E_0}{2C}\left[\cos(kz-\omega t) - \cos(kz+\omega t)\right]\hat{j}$

$\vec{B} = \dfrac{E_0}{C}\hat{j}\sin(kz)\sin\omega t$

$\hat{C} = \hat{E}\times\hat{B}$

$-\hat{k} = \hat{i}\times\hat{B}_1$

$\hat{B}_1 = -\hat{j}$

$\hat{k} = \hat{i}\times\hat{B}_2$

$\hat{B}_2 = \hat{j}$

18. C

Sol.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

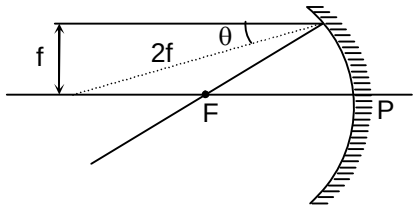
19. B

Sol. Initial phase is $\dfrac{5\pi}{6}$ and $w = \dfrac{\pi}{6}$

$\therefore$ equation: $x = 10\sin\left(\dfrac{\pi t}{6} + \dfrac{5\pi}{6}\right)$

20. B

Sol. $f = \frac{R}{2} \quad \Rightarrow R = 2f$



$\sin\theta = \frac{f}{2f} \qquad \Rightarrow \theta = 30°$ (θ is not small)

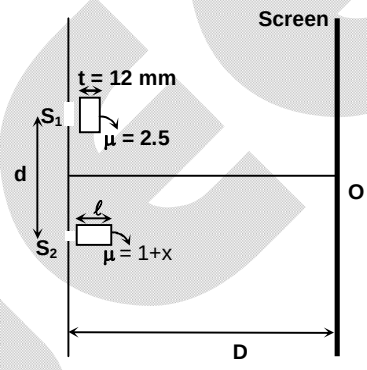
$PF = R - \frac{R}{2}\sec\theta$

$PF = 2f\left(1 - \frac{1}{\sqrt{3}}\right)$

SECTION – B

21. 6

Sol. Optical path length $= \int \mu dx = \int_0^{\ell} (1+x)dx = \ell + \frac{\ell^2}{2}$



For central maxima,

$\Delta x = 0$

$(S_2O - \ell) + \left(\ell + \frac{\ell^2}{2}\right) - \{(S_1O - t) + 2.5t\} = 0$

$\Rightarrow \ell = 6$ mm

22. 4

Sol. $V_A - 2 - 6 + 4 = V_B$

$V_A - V_B = 4V$

23. 1

Sol. $V_1 + V_2 = 6$ …(i)

$P_{MO} = 2P_R$

$(V_{MO})i = 2V_R i$

$V_1 = 2V_2$ …(ii)

From (i) and (ii)

$V_1 = 4, V_2 = 2$

$\sqrt{V_i} = i$

$\Rightarrow i = 2A$

$iR = 2$

$(2)R = 2$

$\Rightarrow R = 1$

24. 750

Sol. $\Delta 102 \approx \Delta 403$

$\frac{10\ Lt}{2P_0} = \frac{V_4 - V_3}{P_0}$

$V_4 - V_3 = 5$ Lt

Work done = ar(Δ102) − ar(Δ034)

$= \frac{1}{2}(V_2 - V_1) \times 2P_0 - \frac{1}{2}(V_4 - V_3) \times P_0$

$= 10P_0 - \frac{1}{2}(5P_0) = 7.5P_0$

$= 7.5 \times 10^5 (pa - Lt) = 7.5 \times 10^5 \times 10^{-3} J$

= 750 J

25. 4

Sol. $\left(\frac{\Delta t}{t}\right) \times 100 = 1.5$

$\Delta t \times 100 = 1.5t$

$E = A^2 e^{-\alpha t}$

$\ell nE = 2\ell nA - \alpha t$

$\left(\frac{\Delta E}{E}\right)\% = \left(\frac{2\Delta A}{A} + \alpha \Delta t\right) \times 100$

$\left(\frac{\Delta E}{E}\right)\% = 2(1.25) + 0.2(1.5t) = 2.5 + 0.2 \times 1.5 \times 5$

$\left(\frac{\Delta E}{E}\right)\% = 4$

26. 8

Sol. $\frac{n\lambda}{2} = L \Rightarrow n(2) = L$ …(i)

(n + 1) (1.6) = L …(ii)

On solving, n = 4 and L = 8 cm

27. 8

Sol. At x = 1m, K.E. is maximum $\Rightarrow v_{max} \Rightarrow a = 0$

$\Rightarrow F - kx = 0$; F = k(1)

At that point, W.E. theorem

$F(1) - \frac{1}{2}k(1)^2 = \frac{1}{2}mv^2 = 4$

$F - \frac{k}{2} = 4$

$F - \frac{F}{2} = 4$

F = 8N

28. 2

Sol. $P = 1.5t^2 = \frac{mdv}{dt}v$

$\int_0^v vdv = \frac{1.5}{2}\int_0^2 t^2 dt$

v = 2 m/s

29. 15

Sol. $v = 3u$

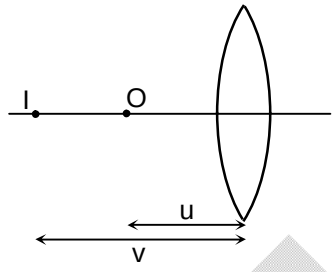
$v - u = 20$

$2u = 20$

$u = 10$ cm

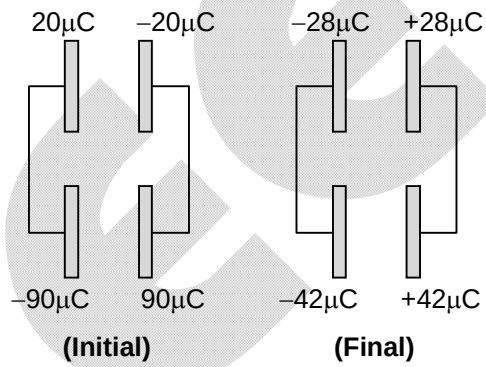
$$\frac{1}{-30} - \frac{1}{-10} = \frac{1}{f}$$

$f = 15$ cm



30. 960

Sol. $\text{Heat loss} = \frac{(48)^2}{2\times 2} + \frac{(48)^2}{2\times 3} = 960$ mJ



Chemistry

PART – B

SECTION – A

31. C

Sol. $r_n \propto \frac{n^2}{Z}$

$\frac{r_{Li^{2+}}}{r_{C^{+5}}} = \frac{3^2}{3} \times \frac{6}{4^2} = \frac{18}{16} = \frac{9}{8}$

32. C

Sol. $Mn^{3+} - d^4$ (low spin) unpaired electrons = 2

$Cr^{3+} - d^3$ (high spin) unpaired electrons = 3

$Co^{3+} - d^6$ (low spin) unpaired electrons = 0

$Fe^{2+} - d^6$ (high spin) unpaired electrons = 4

33. C

Sol. $K_{eq} = \frac{K_f}{K_b}$ $\qquad K_{eq} = \frac{K_f'}{K_b'}$

$K_b' = \frac{4 \times 10^{-6}}{2 \times 10^3}$

$= 2 \times 10^{-9}$

34. B

Sol. 1 and 4 doesn't have plane of symmetry or centre of symmetry.

35. C

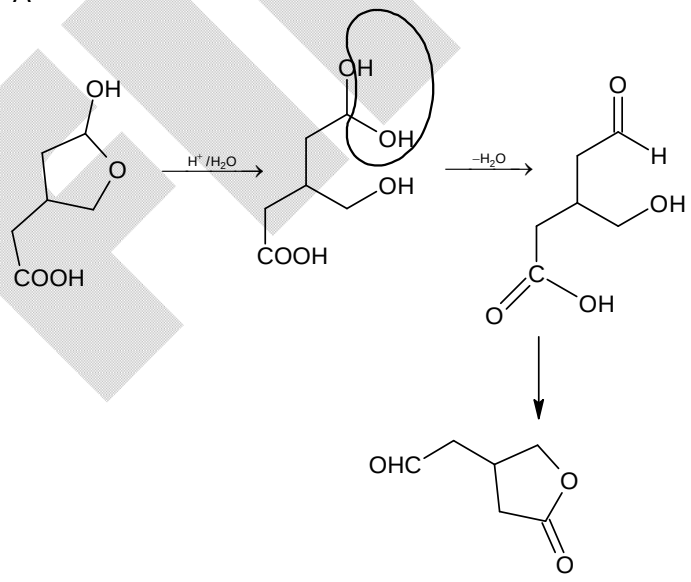
Sol. Electrons of metal centre are transferred to empty π^* orbitals of CO.

36. C

Sol. Due to large size of atoms catenation power decreases.

37. A

Sol.



38. B

Sol. $_{88}Ra^{224} \longrightarrow {}_{87}X^{216} + 2.{}_{2}He^{4} + 3.{}_{-1}\beta^{0}$

39. D

40. A

Sol. $P_4O_6 + 6H_2O \longrightarrow 4H_3PO_3$

$P_4S_{10} + 16H_2O \longrightarrow 4H_3PO_4 + 10H_2S$

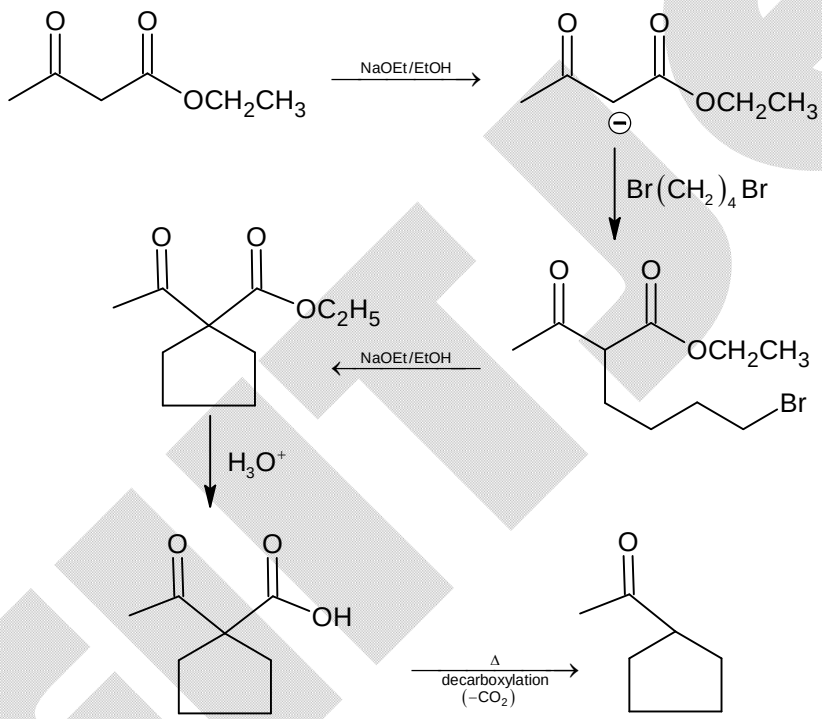
$P_2O_5 + 3H_2O \longrightarrow 2H_3PO_4$

$PCl_5 + 4H_2O \longrightarrow H_3PO_4 + 5HCl$

41. C

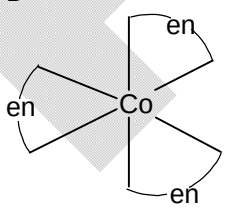
42. C

Sol.



43. D

Sol.



doesn't posseses POS or COS.

44. D

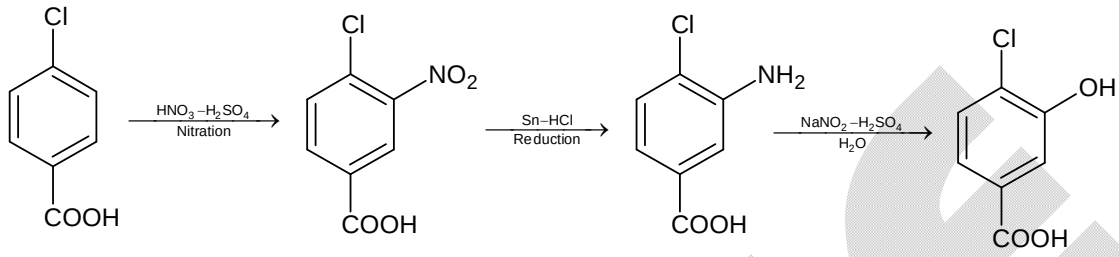
Sol. $\Delta S_{total} = \frac{q_{rev}}{T_h} + \frac{q_{rev}}{T_C}$

$$= \frac{-60}{360} + \frac{+60}{300}$$

$$= -\frac{1}{6} + \frac{1}{5} = \frac{1}{30} = 0.033 \text{ J K}^{-1}$$

45. D

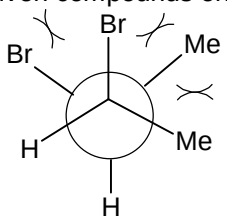
Sol.



46. D

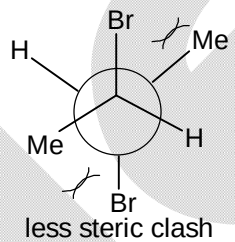
Sol. In the given compounds only (A) and (D) are meso conformers.

(A)



more steric clash

(D)



less steric clash

47. D

Sol. $Na_2\left[Fe(CN)_5 NO\right] \xrightarrow{S^{2-}} Na_4\left[Fe(CN)_5 (NOS)\right]$

Sodium nitroprusside Violet

48. A

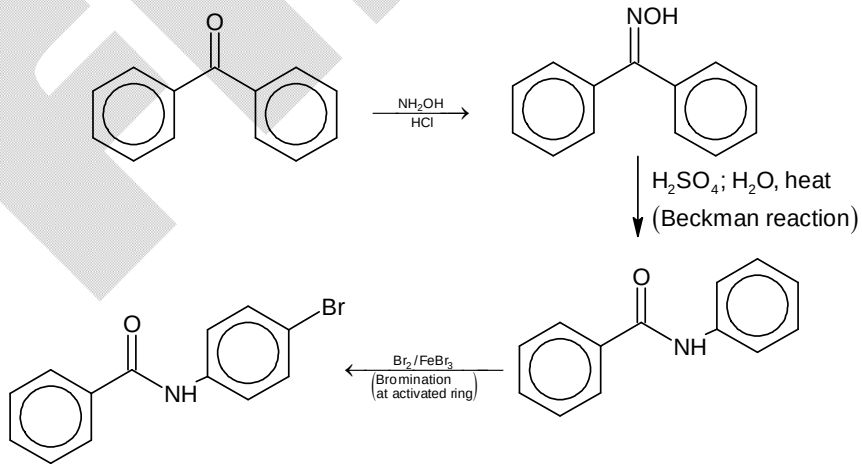
Sol. $HA \rightleftharpoons H^+ + A^-$

$$pH = pK_a + \log\frac{[A]}{[HA]}$$

$$= 4.2 + \log\frac{80}{20} = 4.2 + 0.6 = 4.8$$

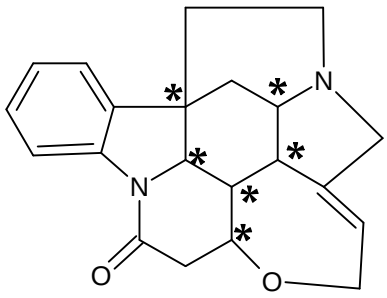
49. C

Sol.



50. A

Sol.



strychnine

SECTION – B

51. 17

Sol. $\frac{n_A}{2} = n_C = \frac{3}{2}n_D$

$\frac{6}{2} = \frac{3}{2}n_D \Rightarrow n_D = 2$

Molar mass $\frac{34}{2} = 17\ g/mol$

52. 8

Sol. $Ti^{3+}\left(d^1\right) - t_{2g}^1 e_g^o$

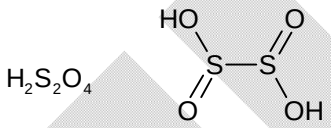
$CFSE = -0.4\Delta_o = -0.4 \times 20{,}000\ cm^{-1}$

$= 8000\ cm^{-1}$

$= 8 \times 10^3\ cm^{-1}$

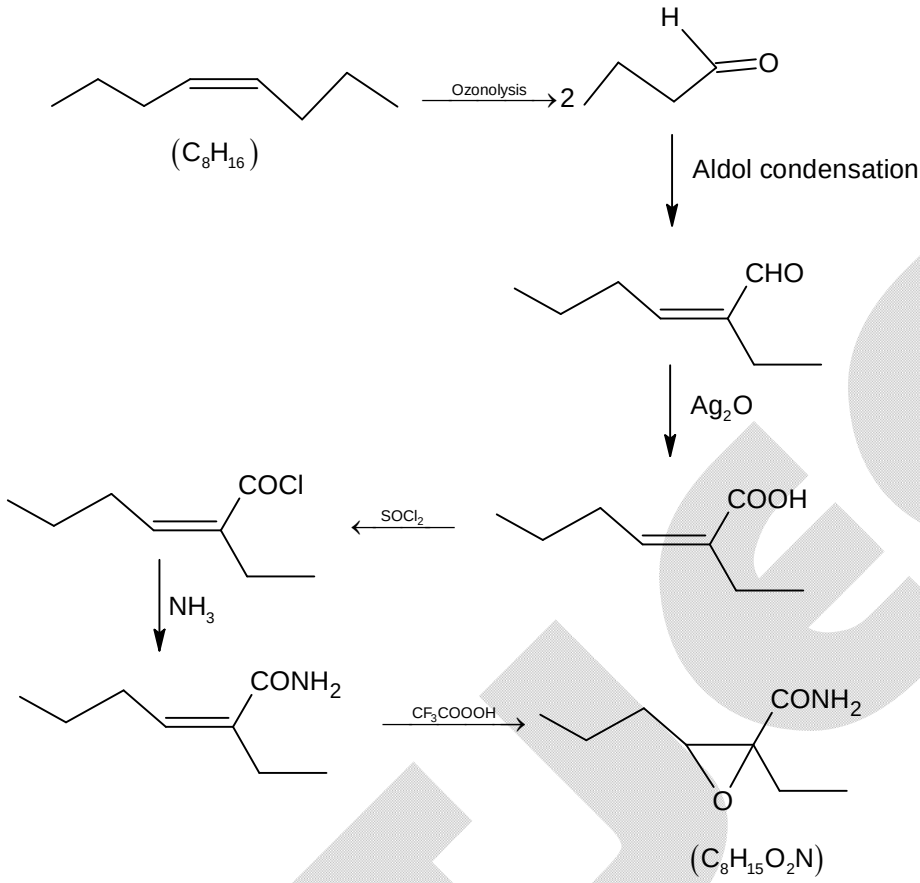
53. 3

Sol.



54. 157

Sol.



Molar mass = 157

55. 405

Sol. $\Lambda_m^o(HCOOH) = \Lambda_m^o(HCOONa) - \frac{1}{2}\Lambda_m^o(Na_2SO_4) + \frac{1}{2}\Lambda_m^o(K_2SO_4) - \Lambda_m^o(KCl) + \Lambda_m^o(HCl)$

$= 105 - \frac{260}{2} + \frac{308}{2} - 150 + 426$

$= 405\ Scm^2\ mol^{-1}$

56. 72

Sol. $H_3C-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3$ (C_5H_{12})

Molar mass = 60 + 12 = 72

57. 20

Sol. $\frac{\pi_A}{C_A.i_A} = \frac{\pi_B}{C_B.i_B}$ (at same temperature)

$\frac{P}{0.01\times 1} = \frac{\pi_B}{0.1\times 2}$

$\pi_B = 20P$

58. 228

Sol. Adding both equation

$$H_2(g)+\frac{1}{2}O_2(g)\longrightarrow H^+(aq.)+OH^-(aq.) \qquad \Delta H^o=-228\ kJ$$

$$-228=0+\Delta H_f^o(OH^-)-0-0$$

$$\Delta H_f^o(OH^-)=-228\ kJ$$

59. 2

Sol. $2XY_2 \rightleftharpoons 2XY + Y_2$

$2(1-\alpha) \quad 2\alpha \quad \alpha$

Total moles = $2+\alpha$

$$K_P=\frac{\left(\frac{2\alpha}{2+\alpha}\right)^2\left(\frac{\alpha}{2+\alpha}\right)}{\left(\frac{2(1-\alpha)}{2+\alpha}\right)^2}\times 4=\frac{\alpha^3 4}{(2+\alpha)(1-\alpha)}$$

$$K_P=\frac{\alpha^3 4}{2}\Rightarrow \alpha=\left(\frac{K_P}{2}\right)^{1/3}$$

$$=\left(\frac{1.6\times10^{-5}}{2}\right)^{1/3}$$

$$=\left(0.8\times10^{-5}\right)^{1/3}$$

$$=\left(8\times10^{-6}\right)^{1/3}$$

$$=2\times10^{-2}\text{ or }2\%$$

60. 12

Sol. Tetrahedral complexes form high spin complexes

Maximum CFSE will be for d^2 and d^7 system

d^2 e_g^2 t_{2g}^o $\qquad$ d^7 e_g^4 t_{2g}^3

CFSE $=-6\times2=-12Dq$ CFSE $=-6\times4+4\times3=-12Dq$

Mathematics

PART – C

SECTION – A

61. C

Sol. $|adj(adj\ A)| = |A|^4 = 6^4$

62. C

Sol. $f(x) = x^2 + \dfrac{1}{x^2}$

63. D

Sol. $L = \dfrac{\pi^2}{2}\left[\int_0^{\pi/2} \dfrac{\sin 2x}{2-\sin^2 2x}dx\right] = \dfrac{\pi^3}{8}$

64. B

Sol. Let $\sec\theta = t$ and integrate to get

$$F(\theta) = -\frac{1}{4}\ln\left(\frac{\sec\theta+1}{\sec\theta-1}\right) - \frac{1}{2}\tan^{-1}(\sec\theta) + \frac{1}{3\sec^3\theta}$$

65. D

Sol. $\because$ ellipse and hyperbola are confocal

Let a_1, b_1 be semi-major axis and semi-minor axis of ellipse respectively and $2a_2$, $2b_2$ be transverse axis and conjugate axis of hyperbola respectively.

$e_1 \rightarrow$ eccentricity of ellipse

$e_2 \rightarrow$ eccentricity of hyperbola

Now, $a_1^2e_1^2 = (13)^2 - (11)^2$

$\Rightarrow\ a_1^2e_1^2 = a_2^2e_2^2 = a_2^2 + 12$

$\Rightarrow\ (PF_1)^2 + (PF_2)^2 = 2\left(a_1^2 + a_2^2\right) = 410$

66. D

Sol. $\left(x^2 + x^4 + \dfrac{1}{x^2} + \dfrac{1}{x^4}\right)^{15} = \left[\left(x^2 + x^4\right)\left(1 + \dfrac{1}{x^6}\right)\right]^{15}$; terms independent of $x = \sum_{r=0}^{5}\left({}^{15}C_{5+r}\right)\left({}^{15}C_{3r}\right)$

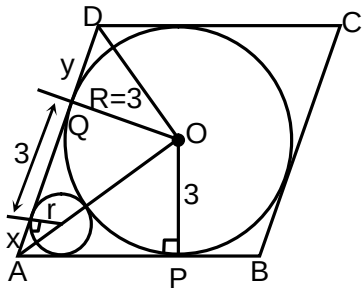
67. A

Sol. $2\sqrt{R \times r} = 3 \Rightarrow r = \dfrac{3}{4}$

$\dfrac{r}{R} = \dfrac{x}{x+3} = \dfrac{1}{4} \Rightarrow x = 1$

$\angle OAP = \dfrac{\theta}{2}$, $\angle ODQ = \dfrac{\pi}{2} - \dfrac{\theta}{2}$

$\therefore\ \dfrac{\tan\theta}{2} = \dfrac{3}{4}$ and $y = \dfrac{9}{4}$



Now ABCD is a rhombus with side $= 4 + \dfrac{9}{4} = \dfrac{25}{4}$

$\because$ Area of ABCD $= \dfrac{75}{2} \Rightarrow d_1d_2 = 75$

$d_1^2 + d_2^2 = 4 \times \left(\frac{25}{4}\right)^2$

$\Rightarrow (d_1 + d_2)^2 = 775$

68. B

Sol. I.F. = $e^{\int \frac{xdx}{x^2-1}} = \sqrt{1-x^2}$

$y = f(x) = \frac{3x^5 + x^4 + 2x^2 + x}{\sqrt{1-x^2}}$

$\int_{-1/2}^{1/2} \frac{f(x)}{(x^2+2)} dx = \frac{\pi}{6} - \frac{1}{4}$

69. D

Sol. L = $\frac{\sum \cos A}{3}$

70. B

Sol. $m \in \left(-\infty, -\frac{4}{3}\right)$ for roots lying in $(-\infty, 1)$ and $3, \infty)$

71. D

Sol. Conceptual

72. B

Sol. $f(x) = x^2 + x$

73. B

Sol. $\sum_{r=1}^{\infty} \tan^{-1}\left(\frac{2r}{1-r^2+r^4}\right) = \sum_{r=1}^{\infty} \tan^{-1}\left(\frac{(r^2+r)-(r^2-r)}{1+(r^2+r)(r^2-r)}\right) = \frac{\pi}{2}$

74. A

Sol. Vertex A is (1, 9)

$\int_1^9 \left(\frac{[x^2]}{[x^2-20x+100]+[x^2]}\right) dx = 4$

75. D

Sol. Conceptual

76. C

Sol. $f(x) = \left(\frac{\pi}{3} - x\right)\sin x$ and $\left(\frac{\pi}{3} - x\right)\sin x < x - \frac{x^3}{6}$ for $x \in \left(\frac{\pi}{3}, \pi\right)$

77. C

Sol. Variable chords pass through a fixed point $(1, -2)$ and locus is $x + y + 1 = 0$.

78. C

Sol. Conceptual

79. C

Sol. $\vec{a}\cdot\left((\vec{a}\times\vec{b})-\vec{b}-c\right)=\vec{a}\cdot(\vec{c}\times\vec{b})-\vec{a}\cdot\vec{b}-\vec{a}\cdot\vec{c}$

$=|\vec{b}|^2-\vec{a}\cdot\vec{b}-3=29-4-3=22$

80. D

Sol. x-coordinate of P = $\dfrac{\frac{r^3}{2}}{r-\sqrt{r^2-\frac{r^4}{4}}}$

$\therefore \quad \alpha=\lim\limits_{x\to0^+}\dfrac{\frac{r^3}{2}}{r-\sqrt{r^2-\frac{r^4}{4}}}=4$

SECTION – B

81. 602

Sol. $Q\equiv(-6a-7,\ 7a+6,\ a)$, $R\equiv(7-2b,\ 2+b,\ 6+b)$

$(-6a-7-\lambda)\times(-6)+(7a+6-\lambda)\times7+(a-\lambda)\times1=0$

$\Rightarrow \quad 43a-\lambda+42=0$ …(1)

$(7-2b-\lambda)\times(-2)+(2+b-\lambda)+(6+b-\lambda)=0$

$\Rightarrow \quad b=1$ …(2)

and $(-6a-7-\lambda)(7-2b-\lambda)+(7a+6-\lambda)(2+b-\lambda)+(a-\lambda)(6+b-\lambda)=0$ ….(3)

using equation (1), (2), (3)

sum of possible values of $\lambda=\dfrac{602}{127}$

82. 73170

Sol. Number of words = $^8C_6\times6!+{}^3C_1\times{}^7C_4\dfrac{6!}{2!}+{}^3C_2\times{}^6C_2\times\dfrac{6!}{2!\times2!}+{}^3C_3\dfrac{6!}{(2!)^3}+{}^7C_3\dfrac{6!}{3!}+{}^2C_1{}^6C_1\dfrac{6!}{3!2!}$

83. 945

Sol. $\sum\limits_{r=1}^{3}|x_r|^2=3\left(|z_1|^2+|z_2|^2+|z_3|^2\right)=945$

84. 1080

Sol. Let $A=\begin{bmatrix}a & b & c\\ d & e & f\\ g & h & i\end{bmatrix}\Rightarrow tr(AA^T)=\Sigma a^2=10$

$\therefore$ Number of matrices = $\dfrac{9!}{6!\times2!}+\dfrac{9!}{2!\times2!\times5!}+\dfrac{9!}{7!}$

85. 729

Sol. $(x-x_1)(x-x_2)\ldots\ldots(x-x_9)=x^9-9$

$k=\dfrac{\left(\frac{9}{x_1}\right)\left(\frac{9}{x_2}\right)\ldots\left(\frac{9}{x_9}\right)}{k}\Rightarrow k^2=9^8$

86. 158

Sol. Common terms of the given 3 AP's are 49 and 109

87. 4

Sol. $(2025)^{3762} = (119 \times 17 + 2)^{3762} = 17\lambda + 2^{3762}$

88. 2447

Sol. Conceptual

89. 31744

Sol. Number of symmetric relations which are not reflexive

$= (2)^{\frac{n(n+1)}{2}} - (2)^{\frac{n(n-1)}{2}}$ (n = 5)

$= 31744$

90. 13

Sol. $\vec{a} \cdot \vec{b} > 0 \Rightarrow \lambda > 0$

$\theta \in \left[0, \frac{\pi}{6}\right) \Rightarrow \frac{\sqrt{3}}{2} < \cos\theta \leq 1$

$\lambda > \sqrt{159}$